Zachary Stone

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Embedded Engineer

Driven embedded engineer seeking to address technological challenges in oceanographic research, contributing to the state of the science and fostering a resilient global ocean.

EDUCATION & ACTIVITIES

B.S. in Electrical Engineering — University of Connecticut, May 2022 — graduated cum laude (3.76)

- Gear Manager, UConn Outing Club — Aug. 2021 – May 2022
- Student Farmer, Spring Valley Student Farm — Aug. 2021 – May 2022

EXPERIENCE

Otis Elevator Company — Farmington, CT

Senior Associate Embedded Software Engineer — Apr. 2025 – Present

Associate Embedded Software Engineer — Apr. 2023 – Apr. 2025

- Develop and maintain real-time, safety-critical motion control software for global high-rise elevator systems in C, built for STM32 ARM Cortex-A microprocessors.
- Created a multi-lingual C#/XAML desktop application for factory configuration of rope integrity monitoring hardware, eliminating a primary source of new equipment field turn-backs.
- Innovated HIL testing by building a C#/XAML desktop application to simulate elevator system CAN communication, enabling motion control software testing on target hardware in a simulated environment.
- Collaborated with field engineering to realize Otis's next generation iPhone-based service tool.
- Redesigned elevator controller software communication interfaces to be independent of platform endianness.
- Led system testing following motion control software port to new computing hardware.

Enviro Power — South Windsor, CT

Electrical Engineering Intern — May 2021 – Sep. 2022

- Designed an ESP32-based data logger PCB utilizing Modbus communication with brownout detection and automatic battery switchover for remote monitoring of boiler health.
- Instituted a PCB quality control procedure and performed quality control assessments on a test stand, reducing unit loss by 8% through hardware fault diagnosis and repair.
- Maintained and improved diagnostic application in C#/XAML for SmartWatt Boiler controller testing.

Power Electronics and Drives Advanced Research Lab — University of Connecticut

Research Assistant — Dec. 2020 – Jun. 2022

- Engineered a variable pole count stator winding for 3 phase induction motors using MotorCAD.
- Secured a $5k grant and built a 3 phase variable-pole induction motor, resulting in a 23% power increase above 2000 RPM compared to a non pole-switching design.

PERSONAL PROJECTS

Audio Equalizer — 2026

- Designed and simulated 3 channel audio equalizer utilizing Sallen-Key active filters with Butterworth filter response.

Slim Wallet — 2025

- Used Python + OpenCV and FreeCAD to design a slim wallet with geometry extracted from images of nature.

Aluminum Hip Belt Buckle — 2024

- Created a novel aluminum hip belt buckle in FreeCAD for laser cut manufacturing + assembly using 3D printed tools.

SKILLS

Languages: ARM Assembly | C | C++ | C# | Python; ***Tools***: FreeCAD | Git | KiCad | Linux | SPICE; ***Embedded***: CAN | HIL testing | I2C | RTOS | SPI | UART; ***Libraries***: cffi | matplotlib | NumPy | pytest; ***Benchtop***: Function Generator | Oscilloscope | Soldering;